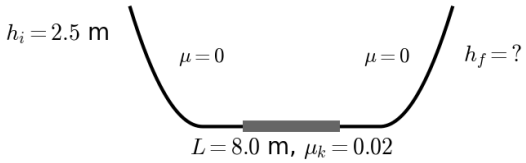
PHYS1408 Practice Exam 2-8

Chapters 6, 7/8, 9, 10, 11

Use $g = 9.8$ m/s^2. Choose the single best answer. Answers are rounded to the precision shown. For a minimum or maximum, choose the answer choice that actually works.

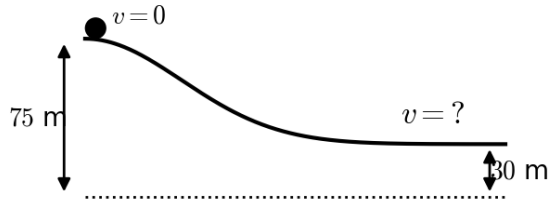
1. A skater starts from rest at a height of 2.5 m on one side of a halfpipe. Overnight a puddle dried and left a rough strip 8.0 m long on the flat bottom with $\mu_k = 0.02$; the rest of the halfpipe is frictionless. How high up the far side does the skater rise?

a) 2.34 m b) 2.26 m c) 2.18 m d) 2.10 m e) 2.02 m



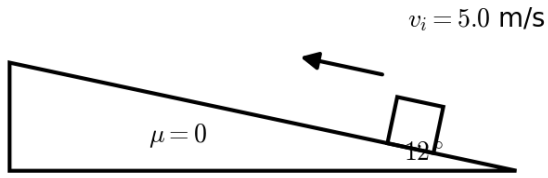
2. A roller coaster car starts from rest at the top of a 75 m hill and rolls down a frictionless track. How fast is it moving at a point on the track 30 m above the ground?

a) 21.0 m/s b) 29.7 m/s c) 38.3 m/s d) 42.0 m/s e) 882 m/s

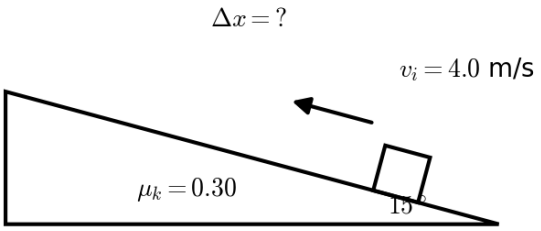


3. A block slides up a frictionless incline ($\theta = 12^\circ$) with an initial speed of 5.0 m/s. How fast is it moving after it has slid 4.0 m along the incline?

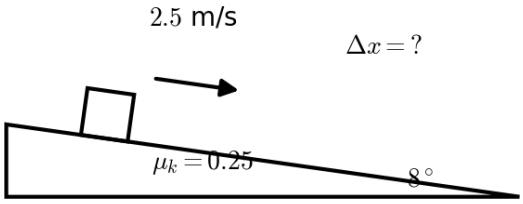
a) 2.95 m/s b) 4.10 m/s c) 6.43 m/s d) 8.70 m/s e) it stops before it slides the whole 4.0 m



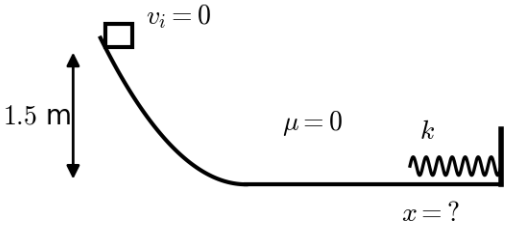
4. A block is given a speed of 4.0 m/s up a 15° incline with $\mu_k = 0.30$. How far up the incline does it slide before it comes to rest?
a) 0.74 m b) 1.49 m c) 2.98 m d) 3.15 m e) 6.30 m



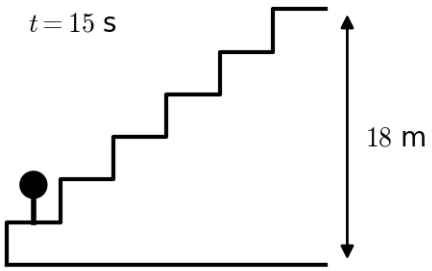
5. A block is sliding **down** an 8° incline ($\mu_k = 0.25$) at 2.5 m/s. How far does it slide before it comes to rest?
a) 0.82 m b) 1.29 m c) 2.94 m d) 5.88 m e) it never stops; it speeds up



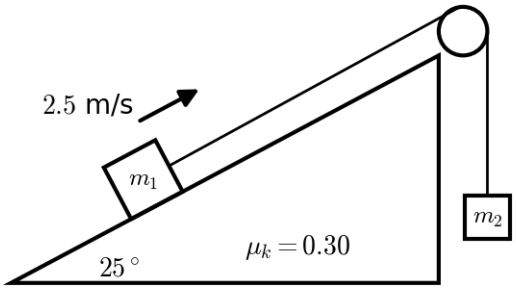
6. A 2.0 kg mass starts from rest at the top of a frictionless ramp 1.5 m high. It slides down and across a frictionless floor into a spring with $k = 1500$ N/m. How far is the spring compressed when the mass comes to rest?
a) 0.020 m b) 0.039 m c) 0.099 m d) 0.14 m e) 0.20 m



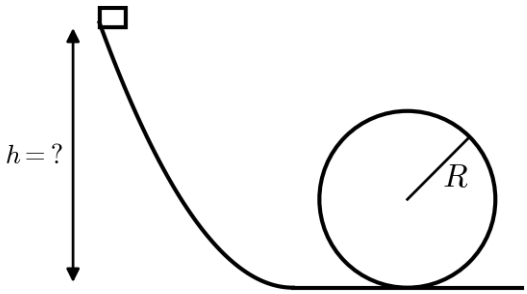
7. An 80 kg man climbs 18 m of stairs in 15 s. What is his average power output?
a) 96 W b) 941 W c) 1880 W d) 14,100 W e) 254,000 W



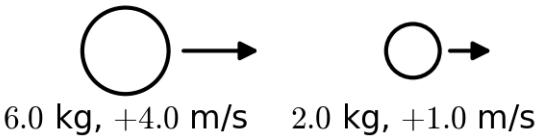
8. A 15 kg block on a 25° incline ($\mu_k = 0.30$) is tied by a light string over a massless pulley to a 3.0 kg hanging mass. The block is moving **up** the incline at 2.5 m/s (the hanging mass is moving down). How fast is the block moving after it has slid 1.0 m up the incline?
a) 1.49 m/s b) 1.62 m/s c) 2.33 m/s d) 3.79 m/s e) it stops before it slides the whole 1.0 m



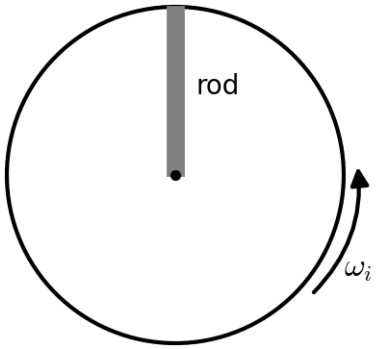
9. A roller coaster car is launched from rest at height h on a frictionless track that leads into a vertical circular loop of radius $R = 9.0$ m. How high must the launch height be for the car to stay on the track (barely) at the top of the loop?
a) 18 m b) 20 m c) 22 m d) 24 m e) 27 m



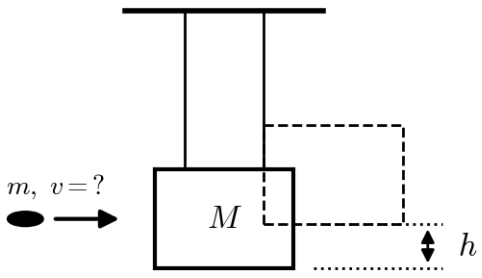
10. A 6.0 kg ball moving right at 4.0 m/s catches up to and collides **elastically** with a 2.0 kg ball moving right at 1.0 m/s. What is the velocity of the 2.0 kg ball after the collision?
a) +1.0 m/s b) +2.5 m/s c) +3.25 m/s d) +4.0 m/s
e) +5.5 m/s



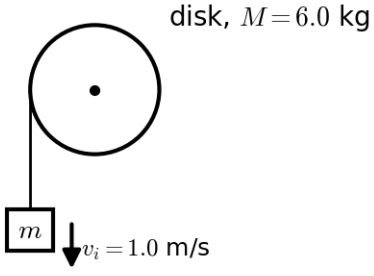
11. A merry-go-round (a disk, $I = \frac{1}{2}MR^2$, $M = 300$ kg, $R = 2.5$ m) rotates at 1.6 rad/s. A 50 kg rod, 2.5 m long, is dropped onto it so that it lies from the center to the edge. What is the new angular velocity?
a) 1.20 rad/s b) 1.37 rad/s c) 1.44 rad/s d) 1.56 rad/s
e) 1.60 rad/s



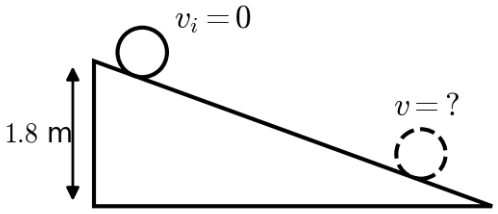
12. A 0.006 kg bullet is fired into a 2.0 kg block hanging from light strings. The bullet lodges in the block, and the block swings up 0.080 m. What was the bullet's speed?
a) 1.25 m/s b) 22.9 m/s c) 334 m/s d) 419 m/s
e) 837 m/s



13. A 2.0 kg mass hangs from a light string wrapped around a pulley that is a uniform disk ($I = \frac{1}{2}MR^2$, $M = 6.0$ kg, $R = 0.40$ m). The mass is moving **down** at 1.0 m/s. How fast is it moving after it has dropped another 1.5 m?
a) 2.62 m/s b) 2.89 m/s c) 3.43 m/s d) 3.57 m/s
e) 5.51 m/s



14. A solid disk ($I = \frac{1}{2}MR^2$) starts from rest and rolls without slipping down an incline from a height of 1.8 m. How fast is its center moving at the bottom?
a) 3.43 m/s b) 4.20 m/s c) 4.85 m/s d) 5.02 m/s
e) 5.94 m/s



15. A disk ($I_1 = \frac{1}{2}MR^2$, $M = 400$ kg, $R = 2.0$ m) rotates at 2.5 rad/s. A second disk ($I_2 = \frac{1}{2}mr^2$, $m = 150$ kg, $r = 1.2$ m) that is **not** rotating drops onto it; it bounces, chatters, and the two stick. What is their final angular velocity?
a) 2.20 rad/s b) 2.50 rad/s c) 2.84 rad/s d) 4.40 rad/s
e) 5.00 rad/s

